Qiyan Ke

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Education

Tsinghua University *Sep. 2025–Present*
M.S. in Computer Application Technology, Department of Computer Science and Technology
Advisor: Prof. Fuchun Sun. **GPA: 4.0/4.0**.

Beihang University *Sep. 2021–Jun. 2025*
B.Eng. in Computer Science and Technology, School of Computer Science and Engineering
GPA: 3.89/4.0; rank: 1/225. Admitted to Tsinghua's master's program via recommendation.
Outstanding Graduate of Beijing. Shen Yuan Medal.
B.S. in Mathematics and Applied Mathematics (Minor Degree), School of Mathematics and Applied Mathematics.

Jiaxiang Foreign Languages School, Chengdu *Sep. 2015–Jun. 2021*
Secondary education. Named one of the school's "Top Ten Students" in 2021.

Research Interests

My research interests are **Reinforcement Learning and Dexterous Manipulation**, with the long-term goal of bringing intelligence into the physical world. I am currently exploring how to improve reinforcement learning with limited samples and how to improve policy performance when the available reward signals are suboptimal.

Honors and Awards

1st Place, 11th Robotic Grasping and Manipulation Competition *ICRA 2026*
Picking from Clutter track. [Official website] [Team video]

National Scholarship *2024*

Shen Yuan Medal, Beihang University *2024*
Beihang's highest student honor; one of 10 undergraduate recipients.

May Fourth Youth Medal, Beihang University *2025*
One of 10 undergraduate recipients.

Computer Science Star, Beihang University *2025*

Beijing Merit Student, Beijing Municipal Education Commission *2024*

Outstanding Graduate, Beijing Municipal Education Commission *2025*

Outstanding Undergraduate Thesis, Beihang School of Mathematics and Applied Mathematics *2025*

Speaking Engagements

Invited Presentation *IROS 2026*
A Multi-Sensor Multi-End-Effector Robotic System for Ordered Picking in Cluttered Bins.
Workshop on Reproducible Benchmarking of Robotic Grasping and Manipulation: From Advanced AI to Generalized Humanoid Intelligence. [Workshop website]

Languages and International Experience

English: CET-6 written score **626**, oral grade **A**. Participated in the 2022 UK summer exchange program (Cambridge, Oxford and Edinburgh), organized by Shen Yuan Honors College, Beihang University.

Research Experience and Publications

Accepted Conference Papers

Minimal-Intervention Target Retrieval in Cluttered Scenes with Grasp-Conditioned Dependency Graphs
CoRL 2026 | Co-first author

Models approach- and lift-stage blockers with grasp-conditioned dependency graphs, combining dependency prediction with short-prefix closed-loop planning to reduce non-target object removals. Constructs approximately 3.66 million object–grasp dependency annotations across 500 scenes, with validation through offline planning, closed-loop MuJoCo simulation, and real-robot experiments.

Ask Professor G. How to Grasp: Training-Free Cross-Gripper Grasping from an RGB-D Observation and a Gripper URDF
IROS 2026 | Co-first author

Combines semantic guidance from foundation models with test-time optimization to generate grasps from an RGB-D observation and a gripper URDF, adapting to new grippers without retraining. Validates cross-gripper generalization and semantically constrained grasping in simulation and on real robots.

Manuscripts Under Review

AIR^2: Repairing Only What Misleads the Policy from Sparse Outcomes
ICLR 2027, under review | First author

Addresses reward-misspecification identifiability under sparse outcome supervision in reinforcement learning. Combines multiscale trajectory-prefix feedback and exploration to repair misleading incentives while preserving useful dense rewards. Jointly estimates reward corrections and a potential function, integrates with BAC and PPO, and evaluates constructed misleading rewards on 22 continuous-control tasks and three public benchmarks.

EAGG: Embodiment-Aligned Grasp Generation via Geometry-Aware Graph Conditioning
IEEE Robotics and Automation Letters (RA-L), under review | Second author

Unifies heterogeneous grippers and dexterous hands through topology-aware end-effector graphs and low-dimensional control spaces, iteratively injecting geometric conditioning during sampling. Achieves 56.17% average success across six training end effectors in MultiGripperGrasp, within 1.10 percentage points of specialized models, and validates fine-tuning and zero-shot transfer.

Evidence-Gated State Regulation for Fault-Tolerant Long-Horizon Ordered Object Retrieval
Science Advances, under review | Co-first author

Validates multimodal physical evidence for state updates and failure recovery to prevent local action failures from propagating through long-horizon tasks. Across 30 paired trials in three cluttered-scene categories, achieves 97.2% target completion and 73.3% full-sequence success, versus 74.7% and 0% for the baseline.

ASTER: Closing the Control Gap via the Agent-State Transactional Execution Runtime
ICLR 2027, under review | Third author

Represents committed agent state as a versioned typed hypergraph, deriving update scope and successor conditions from relation contracts. Verifies model-proposed transitions before atomic commit to preserve unaffected valid state. Across nine external benchmarks spanning embodied execution, multi-agent coordination, service/web interaction, and code generation, improves task success by 3.00–23.61 percentage points over the strongest compared baseline, and supports integration with existing agent harnesses.

Teaching and Student Leadership

Teaching Assistant, Beihang University — *Feb. 2025–Jun. 2025*
Software Engineering.

Teaching Assistant, Beihang University — *Feb. 2025–Jun. 2025*
Introduction to Artificial Intelligence.

Teaching Assistant, Beihang University — *Jul. 2023–Jun. 2024*
Object-Oriented Design and Construction.

President, Student Union, School of Computer Science, Beihang University — *Sep. 2023–Oct. 2024*